Expanded Differentiation Capability of Human Wharton's Jelly Stem Cells Toward Pluripotency: A Systematic Review.

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Human Wharton's jelly stem cells (HWJSC) can be efficiently isolated from the umbilical cord, and numerous reports have demonstrated that these cells can differentiate into several cell lineages. This fact, coupled with the high proliferation potential of HWJSC, makes them a promising source of stem cells for use in tissue engineering and regenerative medicine. However, their real potentiality has not been established to date. In the present study, we carried out a systematic review to determine the multilineage differentiation potential of HWJSC. After a systematic literature search, we selected 32 publications focused on the differentiation potential of these cells. Analysis of these studies showed that HWJSC display expanded differentiation potential toward some cell types corresponding to all three embryonic cell layers (ectodermal, mesodermal, and endodermal), which is consistent with their constitutive expression of key pluripotency markers such as OCT4, SOX2, and NANOG, and the embryonic marker SSEA4. We conclude that HWJSC can be considered cells in an intermediate state between multipotentiality and pluripotentiality, since their proliferation capability is not unlimited and differentiation to all cell types has not been demonstrated thus far. These findings support the clinical use of HWJSC for the treatment of diseases affecting not only mesoderm-type tissues but also other cell lineages. Impact statement Human Wharton's jelly stem cells (HWJSC) are mesenchymal stem cells that are easy to isolate and handle, and that readily proliferate. Their wide range of differentiation capabilities supports the view that these cells can be considered pluripotent. Accordingly, HWJSC are one of the most promising cell sources for clinical applications in advanced therapies.

Recent Patents Pertaining to Immune Modulation and Musculoskeletal Regeneration with Wharton's Jelly Cells.

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Umbilical cord mesenchymal stromal cells (UCMSCs) are isolated from Wharton's jelly in the umbilical cord at birth, and offer advantages over adult mesenchymal stromal cells (MSCs) such as highly efficient isolation, faster proliferation in vitro, a broader differentiation potential, and non-invasive harvesting procedure. Their expansion and differentiation potential renders them a promising cell source for tissue engineering and clinical applications. This review discusses recent updates on the differentiation strategies for musculoskeletal tissue engineering including cartilage, bone, and muscle. In addition to tissue engineering applications, UCMSCs can be utilized to support hematopoiesis and modulate immune response. We review the patents relevant to the application of MSCs including UCMSCs in hematopoiesis and immune modulation. Finally, the current hurdles in the clinical translation of UCMSCs are discussed. During clinical translation, it is critical to develop large-scale manufacturing of UCMSCs as well as the composition of expansion and differentiation media. Four clinical trials to date have examined the safety and efficacy of UCMSCs. Once public banking of UCMSCs is available to supply matched allogeneic units and once UCMSC manufacturing is standardized, we anticipate that UCMSCs will be more widely used in clinical trials.

Epithelial In vitro Differentiation of Mesenchymal Stem Cells.

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Mesenchymal stem cells or mesenchymal stromal cells (MSCs) are non-hematopoietic stromal cells that reside in many human organs and have been isolated from a variety of adult or fetal tissues such as adipose tissue, bone marrow and umbilical cord Wharton's jelly, among others. Because they are a heterogeneous population, International Society for Cellular Therapy has established 3 minimum criteria to characterize MSCs in vitro: i) adherence to plastic, ii) differentiation potential (osteogenic, chondrogenic and adipogenic lineages) and iii) expression of specific surface antigens (CD73+, CD90+, CD105+, CD34-, CD45-, CD11b-, CD14-, CD19-, CD79a-, HLA-DR-). Because of these characteristics, MSCs are useful for different applications and studies, most of them related with regenerative biomedicine. Epithelial in vitro differentiation of MSCs, for clinical use, is one of the main objectives in this field, due to, on the one hand, the difficulties to establish epithelial cell cultures and, on the other hand, the immunomodulatory capacity of MSCs that could increase the success of transplantation. According to this and the information compiled from bibliography, production of epithelial cells differentiated in vitro from MSCs is a complex procedure and a lot of techniques and culture media are necessary to explore. The objective of this review is to show the different methods of epithelial in vitro differentiation and remark the need to further study for being capable of establishing specific cell lines of epithelial cells differentiated from autogenic or allogenic MSCs.

Fetal mesenchymal stem cells in cancer therapy.

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There is compelling evidence that mesenchymal stem cells (MSCs) can be utilized as delivery vehicles for cancer therapeutics. During the last decade, bone marrow MSCs have been used as delivery vehicles for the local production of therapeutic proteins in multiple tumor types, taking advantage of their innate tropism to the tumor site and their low immunogenicity. More recently, MSCs have been isolated from fetal tissues during gestation or after birth. Fetal MSCs derived from amniotic fluid, amniotic membrane, umbilical cord matrix (Wharton's jelly) and umbilical cord blood are more advantageous than adult MSCs, as they can be isolated noninvasively in large numbers without the ethical reservations associated with embryo research. Several studies have documented that fetal MSCs harbor a therapeutic potential in cancer treatment, as they can home to the tumor site and reduce tumor burden. This natural tumor tropism together with their low immunogenicity renders fetal MSCs as powerful therapeutic tools in gene therapy-based cancer therapeutic schemes. This review summarizes various approaches where the tumor-homing capacity of fetal MSCs has been employed for the localized delivery of anti-tumor therapeutic agents.

Umbilical cord mesenchymal stem cells: adjuvants for human cell transplantation.

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The Wharton's jelly of the umbilical cord is rich in mesenchymal stem cells (UC-MSCs) that fulfill the criteria for MSCs. Here we describe a novel, simple method of obtaining and cryopreserving UC-MSCs by extracting the Wharton's jelly from a small piece of cord, followed by mincing the tissue and cryopreserving it in autologous cord plasma to prevent exposure to allogeneic or animal serum. This direct freezing of cord microparticles without previous culture expansion allows the processing and freezing of umbilical cord blood (UCB) and UC-MSCs from the same individual on the same day on arrival in the laboratory. UC-MSCs produce significant concentrations of hematopoietic growth factors in culture and augment hematopoietic colony formation when co-cultured with UCB mononuclear cells. Mice undergoing transplantation with limited numbers of human UCB cells or CD34(+) selected cells demonstrated augmented engraftment when UC-MSCs were co-transplanted. We also explored whether UC-MSCs could be further manipulated by transfection with plasmid-based vectors. Electroporation was used to introduce cDNA and mRNA constructs for GFP into the UC-MSCs. Transfection efficiency was 31% for cDNA and 90% for mRNA. These data show that UC-MSCs represent a reliable, easily accessible, noncontroversial source of MSCs. They can be prepared and cryopreserved under good manufacturing practices (GMP) conditions and are able to enhance human hematopoietic engraftment in SCID mice. Considering their cytokine production and their ability to be easily transfected with plasmid-based vectors, these cells should have broad applicability in human cell-based therapies.

High yield recovery of equine mesenchymal stem cells from umbilical cord matrix/Wharton's jelly using a semi-automated process.

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Umbilical cord is an abundant source of perinatal, plastic adherent mesenchymal stem cells (UC-MSCs). UC-MSCs exhibit robust stemness and strong immunosuppressive and regenerative effects in vivo. This protocol describes enzymatic and mechanical dissociation of umbilical cord matrix (Wharton's jelly) that results in efficient isolation of large numbers of fresh nucleated umbilical cord regenerative cells (UC-RCs) that, when cultured on plastic, exhibit similar characteristics of UC-MSCs. This protocol potentially alleviates the need for culture expansion to obtain large numbers of cells required for clinical application. Dissociation is achieved with a blend of collagenase and neutral proteases with agitation at 37 °C in a semi-automatic system. Average expected yield is $1.65 \times 10(6)$ cells/g tissue with 93 % viability. This protocol has been successfully used to isolate an uncultured nucleated regenerative cell population (also referred to as stromal vascular fraction or SVF) from surgically debrided skin and from human, equine, and canine adipose tissue. The procedure requires less than 30 min for tissue dissection and less than 100 min for cell extraction. Quickly obtaining a large number of UC-RCs that have pluripotent differentiation capacity without the complexity and risks of culture expansion could simplify and expand the use of UC-RCs in clinical as well as research applications.

A Novel Nonhormonal Treatment for Vasomotor Symptoms of Menopause.

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Vasomotor symptoms of menopause, more commonly called hot flashes and night sweats, affect up to 80% of individuals going through the menopausal transition. Hormone therapy with estrogen and often progesterone is the most effective treatment for these symptoms. Many people, however, cannot take estrogen or do not want to take hormones. Many individuals seek nonhormonal, over-the-counter treatment options that have little safety and efficacy information to support their use. In March 2023, the U.S. Food and Drug Administration approved fezolinetant (Veozah), a neurokinin 3 receptor antagonist for the treatment of vasomotor symptoms of menopause. This article presents an overview of fezolinetant, including appropriate usage, adverse effects, its use in special populations, and implications for nursing practice.

Long-term safety of fezolinetant in Chinese women with vasomotor symptoms associated with menopause: the phase 3 open-label MOONLIGHT 3 clinical trial.

PMID: 38818887 | Indexed in PubMed

We aimed to assess long-term safety and tolerability of fezolinetant, a nonhormonal neurokinin 3 receptor antagonist, among Chinese women with vasomotor symptoms associated with menopause participating in the MOONLIGHT 3 trial. In this phase 3 open-label study, women in menopause aged 40-65 years received fezolinetant 30 mg once daily for 52 weeks. The primary endpoint was frequency and severity of treatment-emergent adverse events (TEAEs), assessed at every visit through week 52 and one follow-up visit at week 55. Overall, 150 women were enrolled (mean age, 54 years) and 105 completed treatment. The frequency of TEAEs was 88.7%. Most TEAEs were mild (63.3%) or moderate (22.7%). The most common TEAE was upper respiratory tract infection (16.0%), followed by dizziness, headache, and protein urine present (10.7% each). There was no clinically relevant change (mean ± standard deviation) in endometrial thickness (baseline, 2.95 ± 1.11 mm; week 52, 2.94 ± 1.18 mm). Alanine aminotransferase and/or aspartate aminotransferase levels >3 times the upper limit of normal were reported in 1.4% of women; no Hy's Law cases occurred. Fezolinetant 30 mg once daily was generally safe and well tolerated over a 52-week period among women in China with vasomotor symptoms associated with menopause.ClinicalTrials.gov Identifier: NCT04451226.

Validation and Application of Thresholds to Define Meaningful Change in Vasomotor Symptoms Frequency: Analysis of Pooled SKYLIGHT 1 and 2 Data.

PMID: 38775925 | Indexed in PubMed

Vasomotor symptoms (VMS), the characteristic symptoms of menopausal transition, are often the primary reason women seek treatment. Current treatment options for VMS include fezolinetant, a nonhormonal, selective neurokinin 3 receptor antagonist. This study aimed to define a clinically meaningful threshold for reduction of moderate-to-severe VMS in postmenopausal women treated with fezolinetant and then apply it in a responder analysis of the pooled trial data. This analysis pooled data from two identical phase 3, double-blind, placebo-controlled studies that randomized women with moderate-to-severe VMS to once-daily fezolinetant 30 mg, 45 mg, or placebo (SKYLIGHT 1 and 2). The frequency of VMS was collected daily using an electronic diary. Patients completed the Patient Global Impression of Change in VMS (PGI-C VMS) instrument, which assessed changes in hot flushes/night sweats at weeks 4 and 12 compared with baseline using a seven-point Likert scale. VMS frequency data were anchored to PGI-C VMS data; the anchor level for meaningful within-patient change in PGI-C VMS was "moderately better." In the pooled population (N = 1022), the mean (standard deviation) estimated thresholds for a meaningful within-patient change in moderate-to-severe VMS frequency were - 5.73 (3.47) at week 4 and - 6.20 (5.18) at week 12. Applying the thresholds for meaningful within-patient change to responder analyses ("missing as non-responder" imputation method) indicated a favorable clinical benefit: greater proportions of responders were observed in the fezolinetant 30-mg and 45-mg groups compared with placebo at week 4 (odds ratio range 2.48-2.91; $P < 0.001$) and week 12 (odds ratio range 1.908-2.68; $P < 0.001$). PGI-C VMS is sensitive to change and correlates with VMS frequency: a reduction of approximately six VMS episodes per day from baseline to week 12 was meaningful at the individual patient level. Fezolinetant provides a meaningful clinical benefit for women with moderate-to-severe VMS associated with menopause and represents an important nonhormonal treatment option. NCT04003155 and NCT04003142.

Systematic review and network meta-analysis comparing the efficacy of fezolinetant with hormone and nonhormone therapies for treatment of vasomotor symptoms due to menopause.

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The neurokinin 3 receptor antagonist fezolinetant 45 mg/d significantly reduced frequency/severity of moderate to severe vasomotor symptoms (VMS) of menopause compared with placebo in two phase 3 randomized controlled trials. Its efficacy relative to available therapies is unknown. We conducted a systematic review and Bayesian network meta-analysis to compare efficacy with fezolinetant 45 mg and hormone therapy (HT) and non-HT for VMS in postmenopausal women. Using OvidSP, we systematically searched multiple databases for phase 3 or 4 randomized controlled trials in postmenopausal women with ≥7 moderate to severe VMS per day or ≥50 VMS per week published/presented in English through June 25, 2021. Mean change in frequency and severity of moderate to severe VMS from baseline to week 12 and proportion of women with ≥75% reduction in VMS frequency at week 12 were assessed using fixed-effect models. The network meta-analysis included data from the pooled phase 3 fezolinetant trials plus 23 comparator publications across the outcomes analyzed (frequency, 19 [34 regimens]; severity, 6 [7 regimens]; ≥75% response, 9 [15 regimens]). Changes in VMS frequency did not differ significantly between fezolinetant 45 mg and any of the 27 HT regimens studied. Fezolinetant 45 mg reduced the frequency of moderate to severe VMS events per day significantly more than all non-HTs evaluated: paroxetine 7.5 mg (mean difference [95% credible interval {CrI}], 1.66 [0.63-2.71]), desvenlafaxine 50 to 200 mg (mean differences [95% CrI], 1.12 [0.10-2.13] to 2.16 [0.90-3.40]), and gabapentin ER 1800 mg (mean difference [95% CrI], 1.63 [0.48-2.81]), and significantly more than placebo (mean difference, 2.78 [95% CrI], 1.93-3.62]). Tibolone 2.5 mg (the only HT regimen evaluable for severity) significantly reduced VMS severity compared with fezolinetant 45 mg. Fezolinetant 45 mg significantly reduced VMS severity compared with desvenlafaxine 50 mg and placebo and did not differ significantly from higher desvenlafaxine doses or gabapentin ER 1800 mg. For ≥75% responder rates, fezolinetant 45 mg was less effective than tibolone 2.5 mg (not available in the United States) and conjugated estrogens 0.625 mg/bazedoxifene 20 mg (available only as 0.45 mg/20 mg in the United States), did not differ significantly from other non-HT regimens studied and was superior to desvenlafaxine 50 mg and placebo. The only HT regimens that showed significantly greater efficacy than fezolinetant 45 mg on any of the outcomes analyzed are not available in the United States. Fezolinetant 45 mg once daily was statistically significantly more effective than other non-HTs in reducing the frequency of moderate to severe VMS. These findings may inform decision making with regard to the individualized management of bothersome VMS due to menopause.